**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# LESSON 101 · Lesson 101 · Multicore CPUs and Parallel Systems

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is a core?

Q2. Define concurrency.

Q3. What does Amdahl's law illustrate?

Q4. A 3×4 array of independent cells: how many cells?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

multicore — Processor with multiple CPU cores.

parallelism — Simultaneous processing of independent tasks.

critical path — Longest dependent chain.

Amdahl's law — Speedup bounded by serial portion.

## New learning

Understand — A multicore CPU integrates two or more processing cores, which can execute independent threads concurrently.

Apply — Parallel processing can reduce completion time when tasks can be divided; a serial dependency is a critical-path constraint.

Evaluate — Shared memory bandwidth, locking, thread scheduling and the serial proportion of a program limit speedup. More cores do not guarantee proportional gains.

## Worked example

Eight equal independent jobs are shared across two identical cores. How many ideal waves?

1. Explain the choice — 8 ÷ 2 = 4 equal-duration waves, versus 8 on one core.

# We Do · Practise with guidance

Twelve equal independent jobs run on four cores. Give ideal waves and one practical caveat.

Explain the choice

12 ÷ 4 = 3 waves; shared-memory contention or scheduling overhead can extend time.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

## You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Twelve equal independent jobs run on three identical cores. Compare ideal waves with one core. [6]

### Question 2

Explain two benefits and two limitations of multicore processing for video rendering. [8]

### Question 3

Half a program must run serially; other half parallelised perfectly across four cores. Compute ideal speedup and interpret. [6]

## Exit Ticket · Answer before checking

Four CPU cores always give exactly four times the performance of one core.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. A CPU processing unit able to run a hardware thread or more, depending on design.

Q2. Several tasks progress over overlapping time.

Q3. Serial work limits parallel speedup.

Q4. 12.

### You Do pages · Independent questions

Q1. Three cores 12÷3=4 waves [3], one core12 waves [2], ideal threefold throughput gain [1].

Q2. Divide independent frames/tiles [2], throughput [2], serial dependencies [2], contention/overhead [2].

Q3. Normalised time 0.5+0.5/4=0.625 [3], speedup 1/0.625=1.6 [2], not fourfold due serial work [1].

### Exit Ticket · Model answer

False. Dependencies and overhead reduce gains.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick and rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding